

ballast, annoy operators and cause eye fatigue. With the advent of solid-state technology and advances in fluorescent light science, these lamps now provide higher intensities of cool, white light across the entire spectrum without flickering and shadows. Even dimming of light has now become possible.

Illumination through fibre-optic cable is done by generating light through

an external power unit and supplying it via a flexible, metal-sheathed fibre-optic cable. The intensity can be adjusted at the generator end, giving the operator a high degree of control over the amount and intensity of light distributed on the inspection area. The highest level of optical inspection magnification and clarity is provided by these microscopes and almost all microscopes in use today are of this type.

High-magnification CCD cameras are also used in optical inspection, where the image is projected onto a colour monitor screen. Their advantage is that multiple individuals can view the same image simultaneously. It is also easy to capture digitised images for transmission over the Internet, or for training videos, etc. Either way, proper lighting is critical.

Illumination colour is very critical for some applications, especially for those using machine vision. The special types of coloured lighting used include yellow bulbs for inspecting photo-resist, and ultraviolet light for inspecting the conformal coatings, plating voids and certain other flaws. The wavelength of light used here is not in the range of harmful UV. Overall, the percentage of speciality coloured lighting used in inspection is quite small, and is specific to certain limited applications.

ESD safety

The issue of electrostatic discharge (ESD) safety in illuminated magnification is an important one. The illuminated magnifier used in the microscope is a powerful source of ESD, suspended right above the workpiece. The manufacturers want to protect their products from ESD and ground everything in contact with the operator and the bench-top. The best way to protect products from potential ESD

damage is to ensure that the magnifier used has quality metal parts, or ESD-safe plastic parts.

Diopter

Diopter is a unit of measurement of optical power of a lens, which is equal to the reciprocal of the focal length in metres. For example, a typical 127mm (5-inch) diameter lens may be rated at 3-diopter, with a magnification of 1.75x. The same lens in 5-diopter would have a 2.25x magnification.

A thorough knowledge of available technology for visual inspection, as well as the specific needs of your inspection application, can help you choose the most appropriate inspection equipment for the job.

Minimising operator fatigue—eye fatigue or otherwise—is essential to maintain good inspection procedures, practices and operator effectiveness. Optical inspection equipment is not as simple as it may seem, and ongoing technological innovations continue to improve both the quality and functionality of visual inspection equipment.

Inspection microscopes from some manufacturers

Manufacturer – Vision Engineering

Model – VisionZ

VisionZ combines good HD imag-



Fig. 1: VisionZ microscope

TABLE I
Inspection Magnification
(Land Width)

Land Width or Land Diameter	Magnification Power	
	Inspection Range	Maximum Referee
>1.0 mm (0.0394 in)	1.5x to 3x	4x
>0.5 to ≤1.0 mm (0.0197 to 0.0394 in)	3x to 7.5x	10x
≥0.25 to ≤0.5 mm (0.00984 to 0.0197 in)	7.5x to 10x	20x
<0.25 mm (0.00984 in)	20x	40x

Land Diameter: A portion of a conductive pattern used for the connection and/or attachment of components

TABLE II
Magnification Aid
Applications—Other

Cleanliness (with or without cleaning processes)	Magnification not required, see Note 1
Cleanliness (no-clean processes)	Note 1
Conformal coating/encapsulation	Note 1, 2
Marking	Note 2
Other (component and wire damage, etc)	Note 1

Note 1: Visual inspection may require the use of magnification, e.g., when fine-pitch or high-density assemblies are present, magnification may be needed to determine if contamination affects form, fit or function.

Note 2: If magnification is used, it is limited to 4x maximum.

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